

A Non-collocated Wearable Framework for Back Support Exoskeleton Payload Estimation

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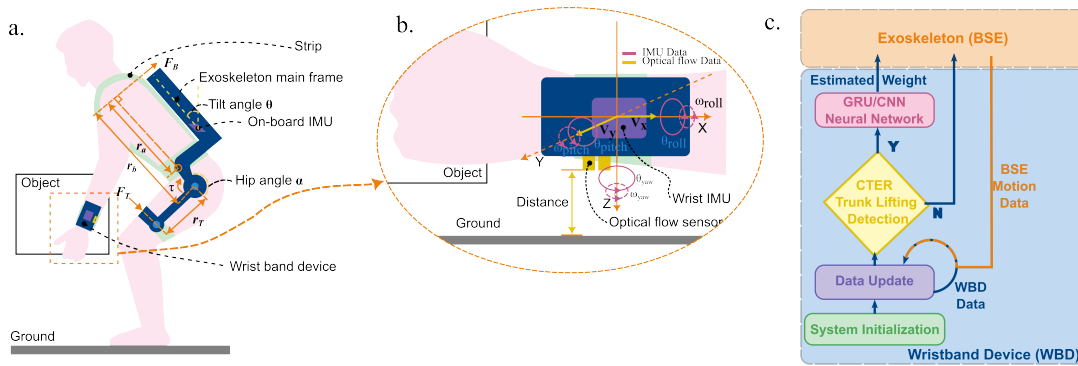


Figure 1: Major Components in Proposed Exoskeleton System (a) Illustration of user fully equipped with the proposed exoskeleton system (b) Wristband device with essential motion data acquired and integrated sensors. (c) System diagram

ABSTRACT

Back-support exoskeletons (BSEs) show promise in preventing occupational low back pain (OLBP). To address a counter-intuitive interpretation of users' demand with the conventional BSEs, which are controlled based on the lifting speed, and offer users to interpret and anticipate the system's output performance. This paper develops a non-collocated, easy-to-equip wearable system enhancing BSEs' capability to perceive operational loads. We designed a Convolution-inspired Trunk Event Recognition (CTER) algorithm for trunk lifting and bending motion detection. Moreover, conceptualizing human motion signals as audio sequences, the study proposed two deep learning classification models leveraging speech processing techniques. This system is capable of detecting body trunk lifting and bending motions, and features onboard estimation of payload categorized into 0, 5, 10, and 15 kg groups.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile computing systems and tools.*



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KEYWORDS

Occupational Low Back Pain; Back Support Exoskeleton; Trunk Motion Detection; Payload Estimation

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1 INTRODUCTION

Up-to-date Occupational low back pain (OLBP) remains a challenge to cure. In response, back support exoskeletons, designed to provide supportive torque during lumbar movements, have emerged as promising tools for mitigating the risk of OLBP[5].

However, for adaptive assistance in exoskeletons, such as adjusting to changes in the user's lifting speed—providing more assistance for faster movements—these methods often overlook the environmental impact on lifting tasks. This oversight can lead to assistance paradoxes, where assistance decreases as the weight of the object being lifted increases due to slower speeds, and vice versa, creating a mismatch between assistance provided and the actual demand.

Therefore, some payload detection systems has been developed to induce load weight information into the BSE system[7, 4]. Inertial

measurement units (IMUs) capture dynamic information, which can be correlated with force-torque parameters to estimate payload weights. This indirect measurement approach is often referred to as non-collocated measurement[7]. Y. Tirupachuri *et al.* introduced a model-driven approach[7] for payload estimation based on 21 IMU sensors. However, complex setup and external computers for computation are needed, which limits its feasibility for onboard computation and industrial deployment. M. Pesenti *et al.*[4] simplified the sensor array to five accessories, and classified weight into four categories (0, 5, 10, and 15 kg). However, their experimental results for onboard applications are not provided.

For trunk lifting and bending motion determination, recent studies have achieved precise movement tracking with complicated measurements[2], and yet no effective algorithm for detecting trunk lifting/bending motion is proposed.

Therefore, we aim to design a non-collocated wearable framework that is compact and easy to wear. The device will leverage the BSE's onboard sensors for a) rapid detection of human trunk lifting motion and, subsequently, b) onboard classification of payload weights. By deploying the Explainable Artificial Intelligence (XAI) models, we expect to enhance transparency and trust for the users when performing various lifting tasks, hence reducing the risk of occupational injuries.

The contributions of this study are:

- (1) Proposed and prototyped a novel framework introducing the perception of in-hand payload weight based on non-collocated human dynamics for back support exoskeletons with only one additional wristband.
- (2) Developed an algorithm for quantitative detection of trunk lifting and bending motions.
- (3) Transferred speech processing techniques to categorize payload weights.

2 METHODOLOGY

2.1 Overall Framework Design

The study incorporates a single wristband device (WBD) in conjunction with the compatible BSE. The WBD integrates the trunk body data acquired by the BSE with the angular and translational dynamics of the wrist joint. Consequently, the proposed framework could capture essential motion data for the study's needs, as depicted in Figure 1(a) and Figure 1(c).

2.2 Back Support Exoskeleton Selection

The WBD is implemented to the back support exoskeleton proposed in our previous work[3]. As shown in Figure 1(a), with BSE onboard sensors, such as the tilt angle of trunk body θ , and hip joint angle respecting ground level α are measured.

2.3 Wristband Sensing System Design

Besides the onboard sensors of the BSE, a single wristband device (WBD), compacted with two types of sensors, is incorporated to capture additional motion data for the users' upper-arm movements. As demoed in Figure 1(b), rotational information (i.e., Euler angles θ_{pitch} , θ_{roll} , θ_{yaw} , and angular velocity ω_{pitch} , ω_{roll} , ω_{yaw})

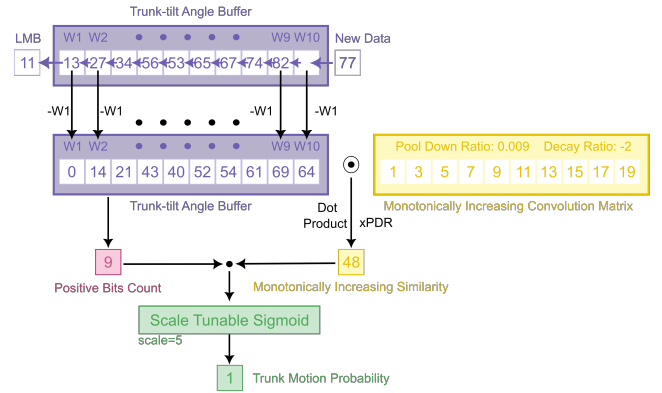


Figure 2: Convolution-inspired Trunk Event Recognition (CTER) Approach

is quantified using a commercial IMU integrated within the wristband. The distance of the wrist joint to the ground D_z , and the relative velocities V_x and V_y are acquired with the near-infrared distance sensor and the single-pixel camera integrated in another optical flow sensor are used to accommodate cumulated noise for translational motion measurement[8].

A microcontroller is implemented to collect the aforementioned parameters from the wristband and wirelessly communicate with the BSE via Bluetooth modules. Attachment is slightly higher from the human wrist joint on the anterior forearm.

2.4 Experiment Setups and Data Collection

A cohort of 7 healthy male participants, where 6 subjects were contributed to the training dataset; 1 "unseen", and 1 from 6 "seen" subjects were invited for real-world validation experiments. Subjects are required to perform stoop lifting tasks on 0 kg, 5 kg, 10 kg, and 15 kg loads with the proposed framework fully equipped.

The sampling for each load group was conducted consecutively in the train set, where onboard validation is shuffled. For each load category, 30 trials were conducted per subject. The subject is instructed when to lift or bend and is required to hold the payload in front of the chest at the end of each trail.

2.5 Human Trunk Lifting Detection

2.5.1 Algorithm Structure. The Convolution-inspired Trunk Event Recognition (CTER) algorithm developed utilizes trunk tilt angle readings from the BSE, and triggers the execution of load weight estimation and the revocation of applied force resistant to bend.

To ensure rapid detection of the trunk motion, a trunk-tilt angle buffer is designed to incorporate new data while simultaneously left-shifting to discard the oldest data. Each bit then subtracts the value of the first position. This adjustment minimizes the influence of varying initial positions or extreme values.

After updating, the buffer is convolved with a tunable, monotonically increasing convolution kernel (MIC). The kernel is defined by two parameters: the pool down ratio (PDR) and the decay ratio, formulated as $MIC = (IdentityFunction) \times (PoolDownRatio) \times (1 - DecayRatio)$. The PDR scales to moderate and prevent extreme values, while the decay ratio adjusts the MIC's slope.

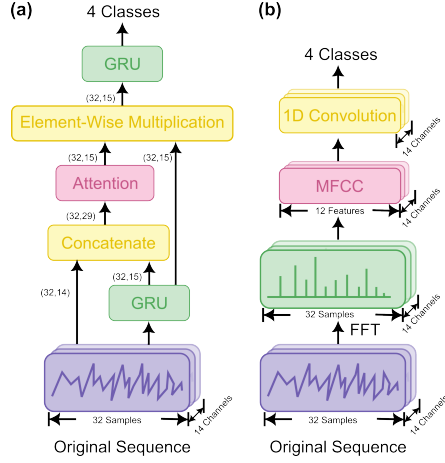


Figure 3: Weight Estimation Network Architecture
(a) GRU with attention mechanism (b) CNN with MFCC feature extraction

The convolution process, where the dot product of the two vectors reflects the phase similarity, offers insight into the correlation between observed and expected trends. The similarity value, derived from the convolution process, is further refined by multiplying it with the positive bit count of the buffer. The positive bit count serves as an indicator of the predominant direction of movement, effectively scaling the similarity value.

Eventually, the yielded value is input to a scale-tunable sigmoid function for the calculation of the trunk lifting motion probability.

2.6 PAYLOAD WEIGHT ESTIMATION

Having successfully achieved automatic trunk lifting detection, this study developed two machine learning models: a) GRU with attention mechanism and b) MFCC feature extraction with Convolutional Neural Network (CNN), for load classification.

For data selection, a comprehensive set of 14 dynamic motion parameters was chosen to inform the payload estimation model: a) BSE: angles and rotation velocities of the hips and trunk tilt angle. b) WBD: Euler angles and velocities, wrist distance to the ground, and relative velocities in the x and y directions.

2.6.1 Problem Formulation. Conceptualize Dynamic Motion as Speech: The data collected during each lifting cycle exhibits a relatively consistent profile, yet the discrepancies indicate that thresholds can be identified through deep learning methods to differentiate between load groups, albeit different subjects.

This scenario can be likened to a sentence expressed in various tones, where the deep learning models in this study aim to discern the underlying 'emotion' (i.e., the payload weight), irrespective of the 'accent' (i.e., variations introduced by different users).

This study addresses a weight classification using dynamic measurements, akin to sentence-level classification. We also hope to enforce the explainability of our models via such processes.

2.6.2 Data Preprocessing.

Method1: GRU with Attention Mechanism: The model's single batch input is a 14-channel time series, each channel representing actual physical measurements with a sequence length of 32 digits.

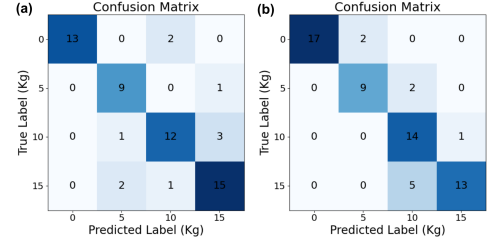


Figure 4: Offline Performance Confusion Matrix of
(a) GRU with attention mechanism (b) MFCC feature extraction with CNN

Table 1: Offline Performance of the Proposed Models

Metric	GRU with Attention	MFCC with CNN
F1 score	0.8321	0.8450
Test Accuracy	0.8644	0.8730
Validation Accuracy	0.8305	0.8413
Precision	0.8392	0.8685
Recall	0.8305	0.8413
Validation Accuracy of Each Class		
0 kg	0.8667	0.8947
5 kg	0.6923	0.6923
10 kg	0.7059	0.6667
15 kg	0.6818	0.6842
Deviation¹	0.0755	0.0930

¹ Deviation between the average validation accuracy

Method2: MFCC Feature Extraction with CNN: The data preprocessing in this method is an extension of the previously mentioned method. Obtaining the 14-channel time series, the Fast Fourier Transformation (FFT) process is executed.

Then, the calculation of the MFCC coefficients follows Equation 1[6], where $c_x[l]$ is the computed l^{th} MFCC coefficient:

$$c_x[l] \triangleq DFT^{-1} \log|\hat{x}[k]| = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \log|\hat{x}[k]| e^{2\pi j k l / N} \quad (1)$$

Subsequently, the dataset was shuffled and divided into training, testing, and validation sets, in 8:1:1 split ratio.

2.6.3 Model Architecture.

Method1: GRU with Attention Mechanism:

The proposed GRU with attention mechanism model is shown as Figure 3(a). The attention mechanism in this study is achieved in the form of sequence attention[1].

$$x_{(t)} = \tanh(W_h \cdot i_{(t)} + b_h) \quad (2)$$

As shown in Equation 2, $i_{(t)}$ is the output from the concatenate layer. The attention score is calculated as Equation 3.

$$\alpha_{(t)} = W_s \cdot x_{(t)} + b_s \quad (3)$$

An element-wise multiplication is implemented to calculate the weighted sum. The final GRU layer classifies the load.

Method2: MFCC Feature Extraction with CNN:

The architecture of the proposed MFCC feature extraction with CNN classification model is as shown in Figure 3(b). The Mel-Frequency Cepstral Coefficients (MFCC) is a widely recognized

Table 2: Onboard Evaluation Results of the Proposed Models

Metric	GRU with Attention	MFCC with CNN
Model Complexity ¹	3365	11914
Prediction Time ²	1.1s	2.254s
Seen Subject Accuracy ³	77.50%	82.50%
Unseen Subject Accuracy ⁴	70.00%	77.50%
Average Accuracy	73.75%	80.00%
Model Size ⁵	40.57 + 11.34 KB	63.09 + 2.38 KB

¹ Number of model parameters

² Measured from data input to the model's output

³ Subject contributed to the training set

⁴ The subject was never invited for data sampling before

⁵ Memory size taken up by the model for storage and execution (Flash + RAM)

speech feature extraction method; then 1D convolution layer is used, followed by a max pooling layer.

2.6.4 Offline Performance. Upon reviewing the validation accuracy for different groups in Table 1 and analyzing the confusion matrix in Figure 4. Method1 exhibits a strong capability of distinguishing between the 0 kg (unloaded status) and non-zero load statuses. This distinction is crucial for recognizing the difference between natural, free-hand human activities and loaded lifting tasks.

As for Method2, despite a slightly higher accuracy deviation among the groups, the predictions predominantly align with the confusion matrix's diagonal line, highlighting the model's precision.

3 RESULTS

3.1 Results of Trunk Lifting Detection

In total, six subjects with 763 real-time trials were successfully recorded. The evaluation metric is accuracy, defined as Equation 4.

$$Accuracy = \frac{TP}{TP + FP + FN} \cdot 100\% \quad (4)$$

In evaluating the CTER algorithm, "True Positive (TP)" refers to successful detection, "False Positive (FP)" to incorrect detection, and "False Negative (FN)" to missed detection. An overall accuracy of 98.29% is reached with an average detection latency of 80 ms.

3.2 Results of Payload Weight Estimation

The weight values were shuffled between each onboard trial. A total of 10 trials were conducted for each weight category.

As shown in Table 2, the Method1's complexity in terms of parameters is smaller than CNNs, emphasizing the model's efficiency. A key metric of interest, the average response time from the start of a lifting action to the output of the estimation result, is recorded at 1.1 seconds. On the other hand, although more parameters are required, the Method2 benefits from the computational efficiency of 1D convolution, necessitating minimal RAM for processing.

An average accuracy of 73.75% and 80.00% for the seen and unseen subjects was observed, respectively. Yet, the significant drop in onboard accuracy and dispersed distribution of confusion matrices compared with the offline training stage could be attributed to the limited training cohort.

4 FUTURE WORK AND LIMITATIONS

Given the current payload estimation time is longer than the duration of human natural lifting tasks, we plan to enhance real-time

performance by a) reducing the complexity of the models by refining the CNN's structure and b) upgrading the embedded processor in the BSEs. The study would expand the dataset to fill the gap between payload estimation accuracy in real-world performance. Meanwhile, by conceptualizing the proposed framework's measurement to speech analysis, only MFCC is deployed; we expect to investigate more time-frequency features in the future.

5 CONCLUSIONS

Aligning the demand of preventing occupational low back pain by enhancing the perception of payload weight in back support exoskeleton systems, this study proposes a simplified wearable framework, consisting of the BSE itself and only one additional accessory – the prototyped wristband device. The framework incorporated the functionalities of (a) human trunk lifting and bending detection and (b) onboard payload weight classification. As part of a proof-of-concept study, despite the existing challenges, the proposed weight estimation method reveals substantial promise for adoption in real-world industrial settings.

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